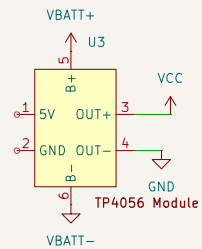
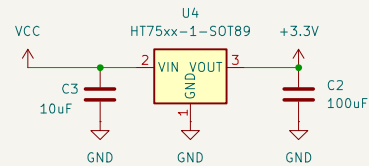


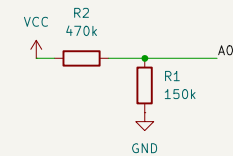
Charging Circuit



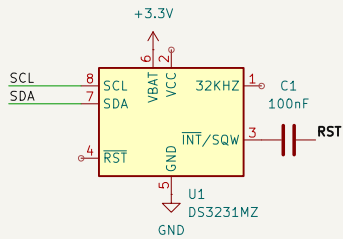
Voltage Regulator



Battery Level Indicator

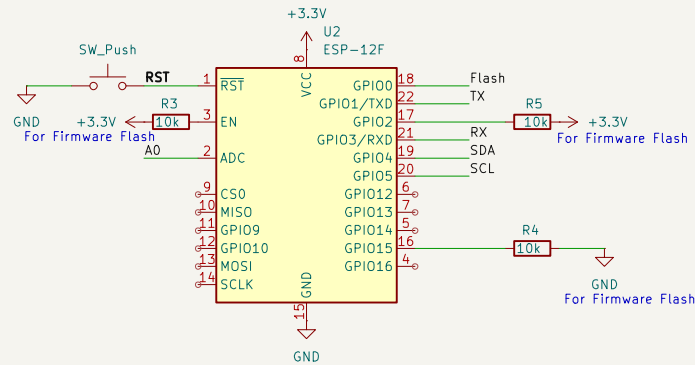


RTC

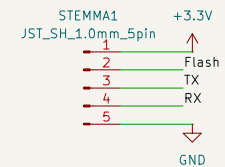


Notes on DS3231:
We should use pullup resistors on SDA, SCL and INT/SQW lines, but in prototype, it worked without the pullups, so omitted here (less power)
When powering the ds3231 for the first time, VCC should be powered. Once the battery powered wakeup register is enabled, you can remove the connection, I guess.

Micro Controller



Stemma Connectors



Change Footprint as required later on. There are various types of STEMMA Connectors, We need the cheap one with 5 pins.